

ISAP — delivering the goods

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Close co-operation between Ericsson and BT has resulted in the rapid industrialisation and deployment of the new interactive speech and applications platform (ISAP), the foundation for a number of potential BT services, the first of which is 'CallMinder', a call answering service. Ericsson's experienced and flexible approach to product manufacture, network installation, operational support and project management were crucial elements in delivering ISAP hardware within the tight time-scales demanded for the introduction of the CallMinder service. The implications of intelligent speech services products like ISAP herald a radical change in key aspects of the telecommunications industry.

1. Introduction

This paper examines a number of issues connected with how Ericsson were able to bring advanced speech technology, pioneered by BT, to the market-place. Although, not academic in the usual sense, there are substantial skills in play and important new lessons that were learnt by all parties. Coverage of ISAP would be incomplete without an insight into this aspect of the life cycle.

Interactive speech services have long been seen as crucial to the development of telephone services. Although speaker-dependent speech recognition has been in commercial use since the 1970s, speaker-independent speech recognition, which can be used to control services for the mass market, has remained something of a holy grail. During the late 1980s, BT were addressing this problem and had developed various demonstration systems where speaker-independent speech recognition was being used successfully. This work identified a need for a platform which could be used to underpin the rapid development and deployment of advanced speech services and led, in turn, to the development of the interactive speech applications platform (ISAP). This platform, however, is much more than a speech recognition device; it is a highly effective, real time computer system of equivalent complexity to a network switch. ISAP combines a wide range of signal and speech processing features in a coherent and integrated manner. It supports a number of alternative network connection and signalling strategies and, as a result, can be used to host a wide range of applications which make use of the specialised speech or signal processing functions to which the platform provides access.

Having a working demonstration is not the same as having a resilient network product. Successful deployment depends on drawing together industrial, technical and commercial skills. Ericsson have extensive industrial experience as developer, manufacturer, supplier and installer of quality network equipment. BT Laboratories (BTL) used their technical expertise to design and develop the pre-production system and write application programs and real time software.

Ericsson's initial role in the project was to ensure the successful transfer of the ISAP into volume production together with the subsequent nationwide roll-out and support of network systems. However, the joint commitment of BT and Ericsson from the outset of the project was very much wider than this. For example, it has always been one of BT's longer term aims to fully transfer ownership of the platform to Ericsson, along with the skills needed for Ericsson independently to continue to develop the ISAP itself and create new and innovative applications. Similarly, it has always been part of the Ericsson strategy to establish ISAP as a successful system within the world market.

In a complex project of this nature involving both practical and commercial considerations, project management skills are key to ensuring a successful outcome, the more so when different companies need to work in close harmony. The project management challenges resulting from this new class of product were substantial. Achieving the short time-scales associated with the transfer

of ISAP into volume production was undoubtedly a direct result of a coherent and co-operative approach to project management spanning Ericsson, BT and a number of the key suppliers.

In addition to the specific involvement with ISAP, the project has highlighted other important aspects for Ericsson:

- ISAP is capable of being used as an 'intelligent peripheral' although not yet fully implemented as such — intelligent peripherals (IPs) as part of an overall intelligent network (IN) will transform the telephony market,
- the industrialisation and deployment of ISAP resulted from an innovative co-operative project between companies; a few years ago analogous arrangements led to major structural changes in the computer industry — given the ever-increasing pace of technological change, it is likely that the telecommunications industry may have to respond in a similar way.

Sections 3 and 4 of this paper summarise the possible implications for Ericsson of these powerful forces.

2. A new challenge

2.1 *How do I order an ISAP?*

Ericsson has well-established processes to manage the delivery of in-house products, for example the AXE10 digital exchange switches. However, at the outset of this project, ISAP was not part of the Ericsson portfolio. A customer could not simply pick up the phone and ask for a quotation. As a result, and given the very tight time-scales which applied, it was clear that the usual processes were unsuitable for building a new and rather different product.

Project management involves a balancing of cost, quality and time. In this case, the short time-scale made it necessary to overlap tasks that would normally be carried out sequentially. This can be a high-risk strategy, potentially threatening quality and cost. In order to minimise this risk, small multi-functional teams were set up by Ericsson; these proved ideal for running the overlapping tasks in a controllable manner. The specific attributes of these teams were:

- team wholly responsible for the project,
- multi-skilled high-quality staff,
- minimal hierarchy and bureaucracy,
- total commitment to the project and each other with no competing work,

- able to take their own decisions and directly experience the consequences.

Throughout the project, the teams comprised a small core of Ericsson staff supplemented by a more fluid number of BT people. During the pre-production phase the team became known as a 'hot-house', underlining the fast growth of a new product. During network roll-out, so-called 'A-Teams' were used, moving rapidly from site to site completing installations and fixing problems.

Despite the very tight deadlines Ericsson delivered the first ISAP system to the network integration facility (NIF) system on time in October 1993. The pre-production system ISAP, based at Ealing and running the pilot CallMinder application entered service on 28 March 1994.

In December 1993 shortly after the manufacture of the NIF pre-production system, BT ordered 21 systems for nationwide deployment. Order placement to acceptance for the 21 systems deployed across the UK was to be just under one year, another very short time-scale. Although, at the time, there was only limited operational experience with ISAP, BT showed considerable confidence in Ericsson and in the commercial prospects for the CallMinder service.

The excellent relationship built up with BT continues as Ericsson has progressively taken over design and development of the platform and the increasingly sophisticated applications required by the market-place. The dispersal of the team into the wider Ericsson and BT organisations has extended ISAP's visibility.

2.2 *Designing the ISAP pre-production system*

BTL started designing the ISAP pre-production system hardware in the early 1990s. The design objectives were to allow easy manufacturing and to minimise lifetime cost. The main design criteria were:

- system architecture based on an industry standard — after some deliberation BT chose the VME bus because of the availability of a wide range of standard hardware,
- platform-specific hardware kept to a minimum — industry expertise used for specialist components, power supplies, backplanes and general-purpose processors,
- system easy to maintain to reduce support costs,
- system components over which BT had direct control were designed to maximise fault detection during production testing, thereby reducing problems during hardware integration.

There were several key design constraints:

- size — the racking was restricted to current exchange equipment practice, leading to a number of compromises, especially on power supply location,

- safety, bump vibration and environment testing — network equipment is subject to particularly severe environmental testing,
- speech processing card — at the time a speech processing card was not available and a specialist sub-contractor was invited to design and build the prototype cards with guidance from BT to ensure that electromagnetic compatibility (EMC), thermal management, manufacturing and reliability standards would be achieved.

Having an average of 200 processors (including digital signal processors) in each ISAP equipment rack led to a number of design issues:

- electromagnetic compatibility — the design contains many more CPUs than an equivalent-sized switching system and therefore, if EMC conformance had not been catered for early in the design, it would have been very difficult to correct at a later stage in the product life cycle,
- power management — modular power supplies allowed for redundancy and variable power requirements (initially, unacceptable voltage transients occurred when powering down a shelf, leading to the remaining shelves restarting; a specialist power supply manufacturer was called in and resolved this problem using a novel approach to capacitor smoothing distributed along the +5 V DC bus),
- thermal management — ISAP requires approximately 100 A of 5 V power per shelf compared, for example, with 50 A for an equivalent-sized AXE 10; the standard solution of three cooling fans per equipment shelf led to dead spots between fans and overheating boards on a full shelf; the eventual solution to this was a five-fan tray with an internal baffle to distribute the airflow evenly.

In order to achieve the tight time-scales for transfer of the ISAP into production, various expedient measures were necessary. For example, Ericsson did not generate its own drawings or detailed construction documents relying instead on those produced by BT, a major saving in time and effort and an example of the high level of trust between the parties.

2.3 The making of ISAP

Design, manufacture and testing

Manufacturing the ISAP, like any other piece of complex computer equipment, is an involved and time critical process. The scope of the manufacturing process extends to obtaining the racks, the power supplies, the

various proprietary circuit cards, etc, together with the production of the ISAP-specific circuit cards, cabling, etc. Ensuring that all the necessary items or components are available at just the right time for completion of the individual sub-assemblies and then for the assembly of the complete system is a vital but often difficult task.

Holding large quantities of potentially expensive components/sub-systems can be very costly; efficient manufacturing processes generally rely on forecasting to procure components in time, a difficult practice for a new product. In addition, to ensure the most effective utilisation of manufacturing capacity, it is usual practice to issue from the stores holding only the exact number of components to complete a specific job.

Experience with ISAP led to the introduction of the 'Kanban' system for low-value common items, where instead of drawing only the right number of components of each kind for a job, the stores issue a 'Kanban' of many components. As many of the components as may be required are used on the job and the remainder returned to the stores holding. As a consequence logistics staff were freed to concentrate on organising and scheduling the delivery of high-value items, like printed circuit boards and VME single-board computer systems used extensively within the ISAP.

While many of the printed circuit cards within the ISAP are standard 'off the shelf' VME processors, two of the processor cards, the signal processor (SP), digital line interface (DLIC) and the line termination (LTC) adaptor are ISAP specific and are manufactured directly by Ericsson. The signal processing card (Fig 1), the most complex of the three and central to ISAP, is an advanced four-level printed circuit card with four digital signal processors (DSPs) and about 400 other components. Components need to be positioned within 0.015 inches, a tolerance more stringent than normal industry practice. In order to manufacture this card, Ericsson acquired a new 'Quad' machine capable of automatically placing components to this 'fine-pitch' close tolerance. The ISAP signal processing card was the first card type produced by Ericsson to use this machine which allowed 70% of the card components to be placed automatically, helping to keep manufacturing costs and time-scales down.

Within manufacturing, a focus on clear definition and documentation of the processes, is key to allowing those activities which impede efficiency to be identified and improved upon. In addition, this clarity of vision is essential if the processes are to be continuously improved, making them faster, more reliable and where possible capable of being done off-line. A good example of this arose with production of the file store systems for the CallMinder ISAPs. Formatting each ISAP disk drive takes approximately one hour and, if it is done 'in-place', has to

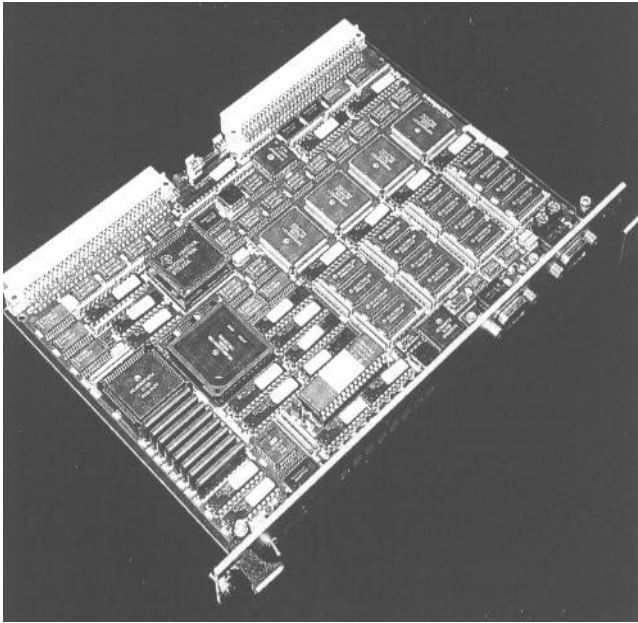


Fig 1 ISAP signal processing card.

be done sequentially, one disk at a time. Since there are 26 disks in each system, formatting 'in-place' holds up production significantly. In order to overcome this problem a piece of equipment that could pre-format 8 disk drives at a time overnight was constructed. Many small changes like this led to an increasing efficiency for all aspects of ISAP manufacture.

Testing is an essential element of the manufacturing process. The objective of the extensive testing programme adopted by Ericsson was not only to improve product quality but also to simplify the building of the product and the running of the tests themselves. Tests which are passed 100% of the time are not worth running and contribute nothing to the quality and reliability of the end product. Within Ericsson, testing is aimed directly at eliminating defects introduced during the manufacturing process. In line with this, the testing regime for every printed circuit board produced for ISAP included:

- checking incoming component quality,
- visual inspection at each stage by the operator and at the end by a trained inspector,
- testing for key functionality in a specially designed jig,
- thermal cycling for eight hours between 0 and 60 °C.

The boards are then assembled into their racks (Figs 2 and 3) and a second series of testing takes place:

- individual boards are tested in the racks,
- hardware is tested in a predetermined sequence,

- initial application tests,
- extensive application testing with a simulated application load.

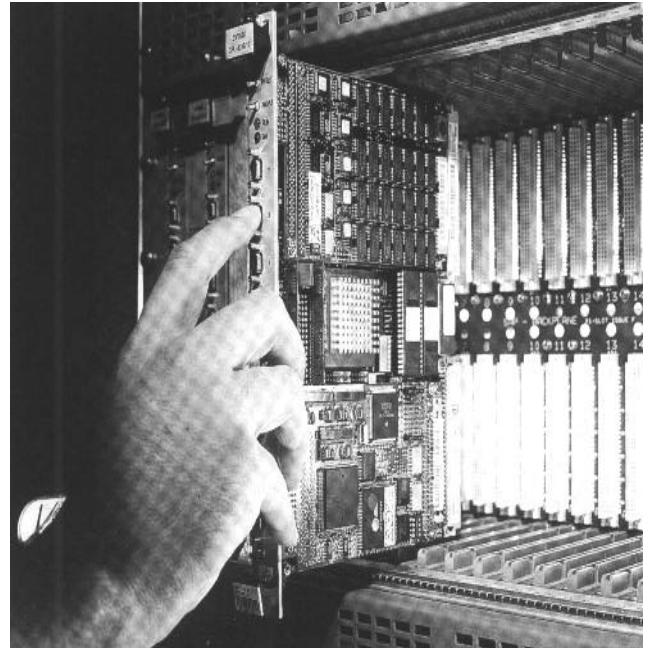


Fig 2 Photograph of inside of ISAP cabinet.

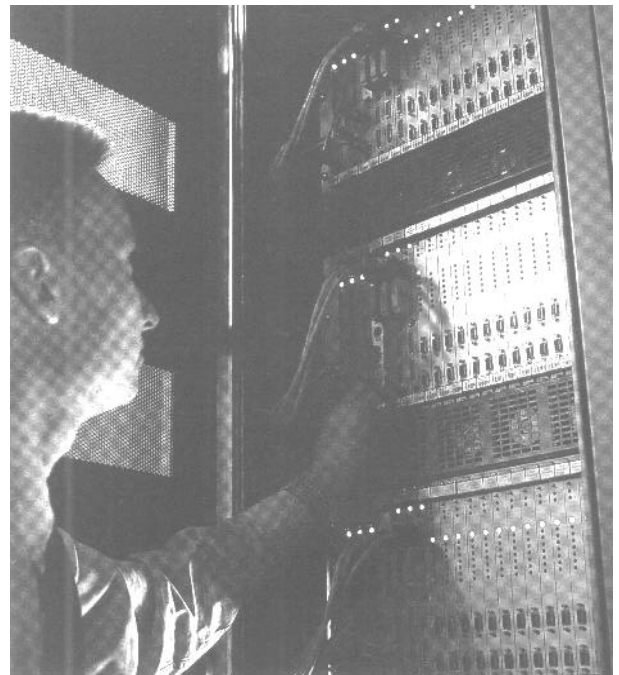


Fig 3 Replacing an ISAP card.

The simulated application load needs to be thorough and easy to administer. Early test scripts for ISAP were controlled by a low-level programming language, time consuming and difficult to use and requiring highly skilled staff. Ericsson subsequently built these into test modules

selected by a higher level language and extended their range and depth. The test operator now assembles a predefined system test on a PC, releasing the skilled computer programmer for other work.

Some initial bump and vibration tests were carried out by BTL on the first ISAP system before installation in the network integration facility (NIF). Subsequently, Ericsson subjected a full system to a formal bump and vibration test as part of network certification. An unscheduled addition to bump testing occurred when a system which was being transported to the United States was dropped during unloading from a lorry. The severity of the damage was such that the individual shelves within the rack were displaced by several millimetres and the plastic handles were sheared off many of the circuit cards. It is a tribute to the design of the platform that the ISAP passed this impromptu test and worked first time.

Network testing on the pre-production system for CallMinder proceeded in stages, a special dispensation from BT for a new product. Lessons learnt during testing were fed into later builds, enhancing product quality. Although small improvements were being made all the time, hardware construction was relatively predictable and 21 CallMinder systems were built in the short period between July and October 1994. Despite the time-scale pressures, no system left the factory without passing all the tests.

Quality management is an essential aspect of modern industrial manufacturing. Ericsson follows the relevant international standard BS5750/ISO9002 which is a continuing demonstration that product quality is being maintained. An important aspect of this standard is the complete documentation of the source and location of every component used. To achieve this, larger components like printed circuit boards are individually bar-coded during assembly. Once the boards are plugged into a rack, a bar-code reader records their exact positions and this information is transferred to a database. This disciplined approach subsequently proved to be of great value when problems were encountered with the quality of one of the components which had been supplied to Ericsson.

Operational support

Operational support covers all aspects of installing the equipment and then keeping both the equipment and service running. In terms of the in-service support, this demands the establishment of a support infrastructure which responds to problems quickly any time of day or night, 365 days/year. This will normally involve first line, second line, third line and design investigation and rectification teams. However, operational support is not just about clearing problems; it is a central tenet of the principles of quality management that, where shortcomings in equipment design or manufacture are identified, corrective action is taken promptly and

effectively. The resolution of two different types of problem encountered by the ISAP systems illustrate this general principle.

- Part of the art of effective commissioning of systems is the ability to deal quickly with the difficulties which, despite extensive planning, inevitably occur. Ericsson had to prepare and put into effect an emergency recovery plan during NIF system commissioning. Problems were encountered when earth connections on the system worked loose. This was potentially a very major set-back for the project, especially as there could have been substantial collateral damage and delay to the overall time-scales. However, a careful check of all boards and replacement earthing screws with plastic locking inserts to ensure that the problem could not recur allowed commissioning to be completed within the original time-scale. Potentially a very damaging incident, but ultimately a very simple and effective cure.
- There had been a small number of system crashes during factory testing which could not be replicated or traced to a specific cause. This was noted, but attributed initially to the early version of application software in use at the time. During network roll-out the frequency of these system crashes increased. Disk read/write errors seemed to be the cause and as the error limits chosen seemed unnecessarily narrow they were relaxed. When the wider limits were also exceeded one of the faulty drives was removed and subjected to a careful test, uncovering a manufacturing fault. The fault was restricted to one batch of disk drives, about 10% of the installed disks. Although, at the time, this was a low point, the wisdom of adhering to quality and configuration management standards soon became apparent. Bar coding meant that it was possible to identify the exact location of each of the faulty drives. Replacement, although an unwelcome distraction, was reduced to a simple engineering change. Without the discipline of configuration management and the bar coding of sub-systems with the associated database, identifying and replacing the affected disks would have been an exceedingly time-consuming and expensive operation.

Technology transfer and evolution of the ISAP platform

As part of the agreement with BT, the ISAP technology is being actively transferred to Ericsson. This involves not just the transfer of drawings and documents, but the active generation of a capability and skill set within the Ericsson organisation. Given the level of understanding and depth of knowledge of the BT development team, which has been gained over several years, this is a significant process. Good progress is being made and Ericsson have now developed their first voice application, independently of BT. Ericsson

are also undertaking a full architectural review of the platform itself such that improvements can be brought to bear in a number of areas, not least the cost.

At a more detailed level, work to improve printed circuit cards continues, for example by reducing the number of components and tuning the design so that more components can be assembled by machine.

In addition, the cabling within the platform is being improved. In the initial designs, interconnections had to be made between connectors mounted at the front of the printed circuit boards and the complexity of the cabling makes both installation and maintenance difficult and error prone. The latest design, which will be in production during 1996, separates the cabling from the boards by providing a row of connectors beneath each row of printed circuit boards. Spare pins on the VME bus provide links between the cards and the new row of connectors. Connecting cables run within this separate section. Connectors are thus eliminated from the front of the boards, allowing boards to be replaced without interfering with any cabling. This will save time during construction and maintenance. It also enhances the integrity of the Faraday cage around the cards giving better EMC performance. Associated changes will allow all maintenance to be carried out from the front, even changing the backplane. This in turn should allow ISAP racks to be installed 'back to back', reducing the floor space requirement (and therefore cost) of an installation.

2.4 Lessons learnt from ISAP

Deploying ISAP, a new kind of element in BT's network has taught both Ericsson and BT a number of valuable lessons. New products of this complexity will always cause some initial difficulties and ISAP with the CallMinder application was no exception. Although this led initially to a disproportionate number of customer support calls the network itself was not affected. The ISAP hardware has been surprisingly robust given its youth and complexity but the application software still has one or two minor shortcomings which Ericsson and BT are working hard to correct.

Competition is increasingly demanding shorter time to market. Lessons learnt in the industrialisation of ISAP include the need to give more attention to:

- initial project funding,
- a dedicated team from the start,
- where possible, earlier contact between designers and the factory to incorporate industrial design expertise to:
 - minimise issues of installation and network integration,
 - enhance modularity giving increased ability to maintain and upgrade,
 - prepare for in service support and documentation,
 - attention to testing, quality assurance and regulatory standards,
 - earlier attention to training.

In retrospect, it is commendable that both companies showed the substantial flexibility that allowed a novel organisational structure to be built around the transfer of the ISAP to Ericsson. Although the ISAP teams are now dispersed into the business, the experience they gained will be invaluable in future projects. The small team approach is now being used for a number of new Ericsson products requiring a short lead time. In the spirit of improving all Ericsson's procedures, it may shortly be desirable to modularise and document this fast-track process.

3. A new market

Through Ericsson's worldwide marketing ISAP has been brought on to a much broader stage, part of Ericsson's worldwide network intelligence portfolio. Applications for ISAP, developed to serve international markets, will also benefit customers in the UK.

ISAP is capable of supporting much more sophisticated services than the CallMinder call-answering service. The ISAP platform already provides true speaker-independent speech recognition, highly accurate for the small vocabulary used by CallMinder. However, the platform's capability is much greater, possibly more than 10 000 words using the advanced 'large recogniser' software for services like automatic operator and directory assistance [1, 2]. A number of other applications for both speech and data services are under development or in operation in BT's network. ISAP is therefore opening up the way for many new voice-operated services.

ISAP is a solutions enabler for PTTs and large private customers. Speech recognition can reduce overheads, increase revenue streams and maintain or improve market share. ISAP provides the capability for a customer to fully interwork with any computer application using voice control alone. As an example, a customer could access a wide range of services through a voice-activated front-end menu. The world-wide development of ISAP is being enhanced through the development of:

- new language support in addition to English,
- enhancement to the signalling stack to cater for additional standards,
- additional applications based on a survey of the requirements of potential customers.

BT have achieved success in seeing their technology taken from the laboratory into a new revenue-generating service. Ericsson have achieved success, not just in industrialising the platform for BT, but also in entering a new market segment. Ericsson will continue to develop ISAP as the world-class product the project team believed from the outset it would prove to be.

4. Conclusions — a view of the future

Change in the telecommunications industry is usually a slow evolutionary process but there are unmistakable signs of an imminent, more rapid, perhaps even revolutionary shift. ISAP breaks much new ground and illustrates many of the forces at work in the industry.

- The CS1 intelligent network interworking standard was not ready at the time ISAP was under development. Nevertheless, the implementation of ISAP has many of the characteristics of an intelligent peripheral in an intelligent network, achieved using C7 signalling. ISAP, therefore, points the way to deployment of other intelligent peripherals in the network. Intelligent peripherals will dramatically change the nature of telephony.
- A major new challenge for the industry is the rapid rise in the complexity of interworking between network elements [3]. Minor interworking restrictions between CallMinder and other services exemplify this point. Industry growth may be constrained if new standards and approaches are not agreed quickly.
- For the most part ISAP uses industry-standard hardware. The key technology is software. Like others in the industry, Ericsson is becoming a systems integrator and software developer and not just a manufacturer of network equipment.
- The kind of support required by BT's customers using sophisticated software applications will become more akin to supporting the small office/home office (SOHO) market. The type and volume of customer support calls on CallMinder underline this trend.
- Product cycles in the computer industry are now heavily overlapped. Cycles are often measured in a few months, whereas a decade ago they were as much as two years. ISAP was on a very short time-scale that

precluded familiar management processes. For competitive reasons time-scales are likely to fall dramatically over the next few years.

- ISAP resulted in part from a commercial licensing agreement between companies, unusual at present in the telephony industry. Inter-company agreements cross-licensing the best technology, have become common in the computer industry and motor industries.
- Ericsson and BT worked together using their complementary expertise to create something neither could have done as effectively alone.

All of these trends have been features of the structural changes in the computer industry during the last decade. The same forces are likely to have a rapid impact on the telecommunications industry. ISAP can thus be seen as a large step in the convergence of the computer and telecommunications industries. Ericsson is addressing the many issues that this raises.

A voice-operated answerphone may seem a small step, but as has been shown the implications are much more significant. As well as providing a service which customers appreciate, ISAP and applications like CallMinder herald significant changes in the telecommunications industry. Telecommunications companies who learn the lessons of the successful computer software players are the likely winners in this rapidly changing industry. At the same time sales of applications built on the ISAP platform are a major opportunity for Ericsson in a new market segment. In this sense ISAP is Ericsson's 'Voice of the Future'.

Acknowledgements

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Phil Twist received an MA degree in electrical sciences from Cambridge University in 1983 and an MBA from Nottingham University in 1990. He has worked for Ericsson Ltd since 1984 and was originally involved in the development of customer premises equipment. After a series of posts in manufacturing, logistics and planning he formed a business unit to hothouse ISAP from pilot into a fully commercialised product. He now heads the marketing function for a number of developing technologies, of which ISAP is a major element.



Terry Briggs has been with Ericsson for nineteen years, and has principally been involved in the telecommunications manufacturing sector. He qualified in Electronic Production Engineering in 1976. Over the years he has been responsible for several major production plant programme implementations to support BT AXE 10 service deployment and the introduction of PCN digital radio into the UK market.

He joined the Public Systems Division of Ericsson in 1993 as the ISAP Operations Manager, supporting programme management for the BT CallMinder service. He was instrumental in introducing the ISAP technology into a volume manufacturing situation with BT Laboratories. Presently, he is heading ETL's export programme with the aim of providing a new Local Exchange Carrier Network in the Philippines.



Kerry Morgan joined Ericsson Public Systems in 1993 as a Senior Project Manager where he was responsible for the CallMinder project during London and national roll-out. He also played an instrumental roll in creating the ISAP organisational infrastructure at Ericsson.

He qualified as a Mechanical, Production and Design Engineer in 1983 and has completed several formal Industrial Management courses. Before joining Ericsson he headed large strategic broadband research projects in co-operation with the CEC. He spearheaded the highly successful RACE project 'CAR' in which BT Laboratories played a key role. He is currently responsible for Global ISAP and Broadband Programmes and has recently taken the technology transfer responsibility for ISAP between Ericsson and BT Laboratories.



Pat Hughes graduated from Liverpool University with a BEng in electronic engineering in 1978 and went on to study for a PhD. Some six years later, having completed his doctorate in digital signal processing and spending three years as a lecturer, he joined BTL, working on speech analysis and synthesis. He became involved in the development of speech platforms working initially with network-based message announcement systems. Later, he worked on the initial design of the SAP and then co-ordinated its onward development. He is now in the Corporate Programme

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